

Signal-Driven Graph RAG for Domain QA: Design, Deployment, and Evaluation

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Abstract—Can a small, locally-deployed language model match a commercial large model on domain-specific question answering when supported by well-designed retrieval? We investigate this question through a signal-driven Graph RAG architecture for Traditional Chinese Bible QA, integrating PostgreSQL, Qdrant, and Neo4j with a 6-route retrieval system that selects optimal engine combinations based on six query signal features. On a 100-question benchmark spanning five question types and 19 metrics, Gemma 3 4B (local, zero API cost) achieves comparable performance to Claude Haiku 4.5 (commercial API): 9 of 12 generation metrics differ by less than 4 percentage points, with complementary strengths—Claude leads on faithfulness (0.887 vs. 0.849) while Gemma leads on answer relevancy (0.658 vs. 0.592). Our results demonstrate that retrieval architecture design, rather than model scale, is the primary determinant of domain QA quality, simultaneously reducing cost and compensating for the limited domain knowledge of smaller models.

I. INTRODUCTION

Large language models (LLMs) achieve impressive performance on general question answering, yet deploying them for domain-specific tasks presents two challenges: (1) commercial API costs scale linearly with query volume—making small, locally-deployed models attractive—and (2) non-English specialized corpora such as Traditional Chinese biblical texts receive limited coverage during pre-training, leaving both large and small models with factual gaps that cannot be resolved by scaling alone. This raises a practical question: *can a well-designed retrieval pipeline close the quality gap between a 4B-parameter local model and a commercial large model?*

Retrieval-Augmented Generation (RAG) [1] reduces hallucination by grounding generation in retrieved evidence, while graph RAG [2] further exploits entity relationships beyond dense retrieval [9]. Yet existing surveys [3] compare strategies under a single model, leaving a gap: *when retrieval is held constant, how much does model scale affect quality?*

Our contributions are threefold:

(1) A signal-driven 6-route Graph RAG integrating PostgreSQL, Qdrant, and Neo4j with decision-tree routing on six query signals; (2) empirical evidence that Gemma 3 4B [6] (local, zero cost) matches Claude Haiku 4.5 [7] with complementary strengths across 12 metrics; (3) a 19-metric evaluation framework [5] over 100 questions in 5 types.

II. SYSTEM ARCHITECTURE

A. Data Processing Pipeline

The 66 books of the Traditional Chinese Bible are processed through hierarchical chunking: Book (66) → Chapter

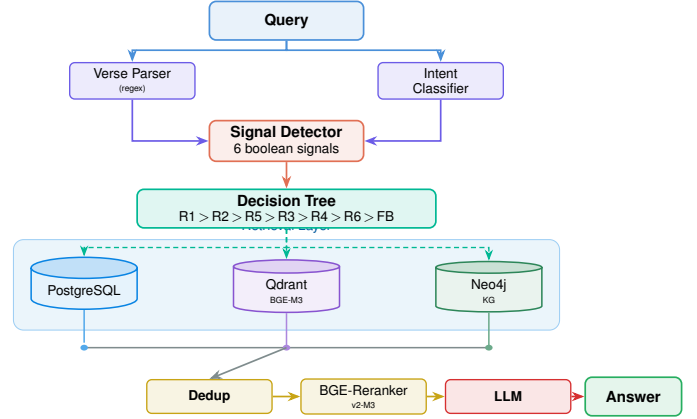


Fig. 1. Signal-driven 6-route RAG pipeline.

(1,189) → Pericope (2,779) → Chunk (438), with token-aware splitting (512–1024 tokens) respecting verse boundaries. CKIP Transformers [8] perform Chinese NER with LLM validation, extracting 14,845 entities across 6 types (Person, Place, Event, Theme, Object, Group) linked to text via 49,668 MENTIONS edges. BGE-M3 [4] generates 1024-dim embeddings for 3,041 vectors stored in Qdrant.

B. Triple-Database Architecture

PostgreSQL stores structured verse/chapter/book data (100,222 records) for exact lookup. Qdrant provides dense vector search via cosine similarity over 3,041 embedding points. Neo4j maintains a knowledge graph with 19,317 nodes and 58,031 relationships across 5 edge types: MENTIONS (49,668), CONTAINS (4,406), NEXT (2,980), CROSS_REFERENCES (912), and NEXT_BOOK (65).

C. Signal-Driven 6-Route Retrieval

Each query undergoes regex verse parsing and LLM intent classification. A *signal detector* evaluates six boolean features—verse reference, chapter reference, multi-book, multi-person, event keyword, and place name—via regex and dictionary matching against knowledge graph entity lists. A priority-ordered decision tree (Table I) routes queries: R1 performs direct SQL verse lookup; R2 combines SQL chapter filtering with semantic search; R3/R4/R6 launch parallel graph traversals (Person/Event/Place→MENTIONS) alongside semantic and SQL retrieval; R5 uses semantic seed passages to feed CROSS_REFERENCES traversal; the Fallback defaults to pure semantic search. All routes except R1 undergo deduplication and BGE-Reranker-v2-M3 [4] reranking.

A pluggable LLM factory supports multiple providers with identical evidence-first prompting; the LLM also performs intent classification, so model switching can affect routing (Section IV).

TABLE I
SIGNAL-DRIVEN ROUTE OVERVIEW AND ENGINE WEIGHT MATRIX

Route	Signal	SQL	Sem.	Graph	XRef
R1	book+ch:v	1.00	—	—	—
R2	book+ch	0.90	0.60	—	—
R3	≥ 2 persons	0.50	0.70	0.90	—
R4	event kw	0.50	0.70	0.85	—
R5	≥ 2 books/xref	0.40	0.65	0.75	0.85
R6	place	0.50	0.70	0.85	—
FB	(else)	—	1.00	—	—

III. EXPERIMENTS

Setup. We curated 100 questions across 5 types (20 each): VERSE_LOOKUP, TOPIC, PERSON, EVENT, and GENERAL, mapped to routes R1–R5 respectively. We compare Claude Haiku 4.5 (commercial API) [7] against Gemma 3 4B (local Ollama) [6] sharing the same retrieval codebase (temperature=0.1, top_k=5). We employ 19 metrics in 4 categories: retrieval (7), reference-based (5), semantic (1), and LLM-judge (6, via RAGAS [5]).

TABLE II
OVERALL PERFORMANCE COMPARISON (SELECTED METRICS)

Category	Metric	Claude	Gemma	Δ
Retrieval	Hit Rate	0.790	0.820	+0.030
	Recall@5	0.746	0.761	+0.015
	MRR	0.753	0.782	+0.029
LLM Judge	Faithfulness	0.887	0.849	−0.039
	Ans. Relevancy	0.592	0.658	+0.066
	Ctx. Recall	0.678	0.712	+0.034
	Ans. Correctness	0.572	0.557	−0.015
Semantic	Similarity	0.738	0.729	−0.009
Ref-based	BERTScore	0.663	0.685	+0.022

TABLE III
ANSWER CORRECTNESS BY QUESTION TYPE (ROUTE)

Type (Route)	Claude	Gemma	Δ
VERSE (R1)	0.841	0.886	+0.044
TOPIC (R2)	0.657	0.622	−0.036
PERSON (R3)	0.440	0.418	−0.022
EVENT (R4)	0.457	0.440	−0.017
GENERAL (R5)	0.416	0.370	−0.047

IV. RESULTS AND DISCUSSION

Finding 1: Complementary Model Strengths. Table II shows that 9 of 12 generation metrics differ by less than 4 percentage points. The models exhibit complementary strengths: Claude achieves higher faithfulness (0.887 vs. 0.849), indicating stricter citation behavior, while Gemma leads on answer relevancy (0.658 vs. 0.592), suggesting more comprehensive responses. Retrieval metrics are identical for signal-deterministic routes (R1, R2) but diverge slightly for R3–R5, where LLM-dependent intent classification affects entity extraction and

thus graph traversal inputs. This confirms that when high-quality context is provided, model differences manifest as generation style rather than fundamental quality gaps.

Finding 2: Route-Specific Analysis. Table III reveals clear route-dependent patterns. VERSE (R1) scores highest (0.841/0.886) because SQL direct match provides exact evidence; Gemma’s stronger BLEU (0.547 vs. 0.173) on this route suggests closer lexical alignment with reference answers. GENERAL (R5) scores lowest (0.416/0.370) as the cross-reference path—the most complex 4-engine route—faces sparser CROSS_REFERENCES edges (912 total). EVENT (R4) also challenges both models (0.457/0.440) since multi-chapter narratives strain fixed-window chunking. Claude leads on TOPIC and GENERAL; Gemma leads on VERSE, indicating complementary biases tied to route complexity.

Finding 3: Cost-Quality Tradeoff. Gemma 3 4B runs locally with zero API cost and full data privacy. Neither model dominates overall, making the local option viable for production. The key investment shifts from model scale to retrieval infrastructure: RAG compensates for smaller models’ limited domain knowledge by supplying structured evidence. Low ROUGE (<0.03) with moderate semantic similarity (0.738/0.729) confirms paraphrased but accurate generation driven by retrieval quality.

V. CONCLUSION

We presented a signal-driven Graph RAG for Traditional Chinese Bible QA with a 6-route decision-tree router over PostgreSQL, Qdrant, and Neo4j. A local Gemma 3 4B matches Claude Haiku 4.5 on 9 of 12 metrics within 4 percentage points, confirming that retrieval design—not model scale—primarily determines domain QA quality while eliminating API costs. Future work includes hybrid dense-sparse retrieval [10], adaptive route weights, and multilingual evaluation.

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